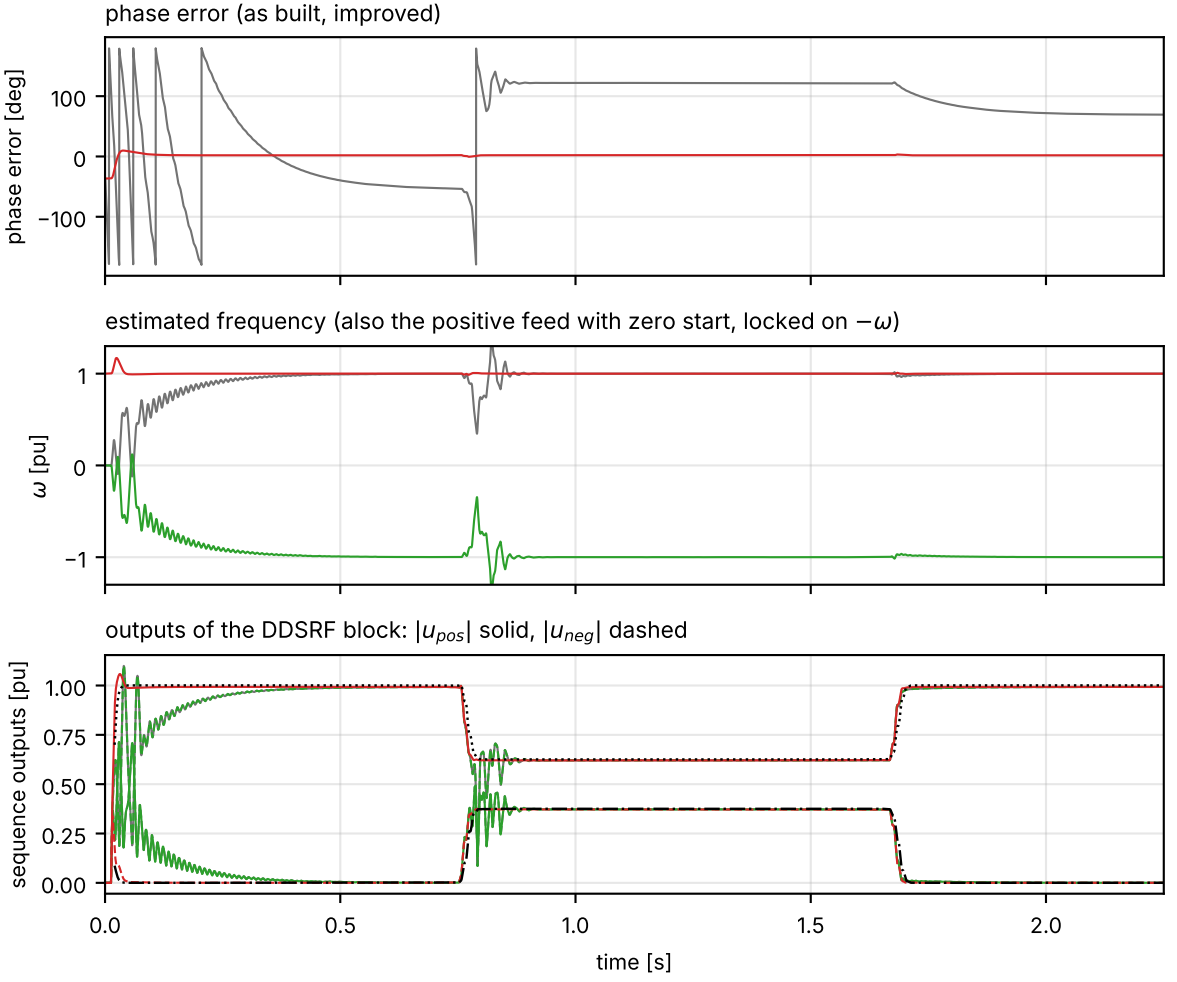
phase error (as built, improved)
phase error [deg]
100
0
−100
estimated frequency (also the positive feed with zero start, locked on −ω)
ω [pu]
1
0
−1
outputs of the DDSRF block: |u_pos| solid, |u_neg| dashed
sequence outputs [pu]
1.00
0.75
0.50
0.25
0.00
0.0
0.5
1.0
1.5
2.0
time [s]
DDSRF-PLL as built (negative feed)
DDSRF-PLL, positive feed (k_p 1, k_i 80)
DDSRF-PLL improved (k_p 0.32, k_i 8, ω init 1 pu)